Drill Builder — PRD

10k Athletics — Team Software & Creator Marketplace

Owner: Blake Wellington · Status: v1 scoped, in roadmap · Last updated: April 2026

Problem

Coaches in the Regis pilot have asked, repeatedly and unprompted, for a way to design and share visual drill diagrams. Today, soccer coaches build practice plans by sketching on whiteboards, drawing in Notes, or pulling images off Google. None of that survives the handoff to players: a coach's diagram lives on a phone screenshot, not inside the IDP a player actually trains against.

This is a real pain point in the practice-planning workflow, and it sits at the intersection of all three 10k services:

- **Player app:** assigned drills today are text + video. A diagram alongside the description is the difference between "run a 4v2 rondo" and a player understanding the spacing without a coach present.
- **Team software:** coaches plan sessions and need to communicate them to the staff and to players. Diagrams are the lingua franca of soccer coaching.
- **Creator marketplace:** a creator like Sheldon sells programs that include drill packs. Right now those packs are video + PDF. Native, structured drill content is the long-term marketplace primitive — it's searchable, remixable, and platform-locked in a way that PDFs aren't.

The marketplace cut is where the strategic value sits. A diagram tool that lives only inside team software is a feature; a diagram tool whose output is a first-class marketplace asset is a moat.

Users & Jobs

User	Job to be done	v1 priority
Coach (HS / club)	Build a session plan with diagrams I can share with my staff and players in under 10 minutes.	Primary. The pilot is built around this user.
Player	Open my IDP for the day, see the drill, understand the shape and spacing without asking a coach.	Primary (read-only). v1 ships rendering before authoring.
Marketplace creator	Build a drill pack once and sell it as native content, not a PDF.	Secondary (v1.1). Creator authoring uses the same tool but ships behind the team-software release.
Assistant coach / staff	Receive a session plan, run it on the field without re-interpreting.	Inherits coach view. No separate flow in v1.

Scope & Approach

Three paths were considered:

- **Custom full build (canvas + animation timeline + export pipeline + sport-specific templates).** Full control, deepest marketplace integration, longest timeline. ~1 quarter for a credible v1.
- **Third-party embed (existing soccer-diagram tool wrapped in our UI).** Fastest to ship, but dependency risk on a small vendor and weak marketplace integration — their export format becomes our content format, which is the wrong direction.
- **Hybrid: ship a static diagram builder first, defer animation.** Recommended. Preserves marketplace optionality (drills as native, structured content), keeps v1 honest, and gives us a real artifact to test with creators before committing engineering to the animation layer.

The deciding question was not "which path is technically best?" but "which path lets us learn whether drills are a marketplace primitive before we spend a quarter building animation?" The hybrid path is the only one that answers that question inside this quarter.

In scope (v1)

- Static drill canvas with a soccer field background (full pitch, half pitch, attacking third presets).
- Drag-and-drop palette: players (light/dark, GK), ball, cones, mannequins, goals.
- Annotation primitives: arrows (run, pass, dribble — three line styles), zones, text labels.
- Save, name, and tag a drill. Tags drive search and reuse.
- Render in the player app's drill view (read-only image at v1; interactive view in v1.2).
- Export to PNG and PDF for coaches who still need to print or share outside the platform.
- Drill library scoped to the coach's team. Personal drafts vs. team-shared distinction.

Out of scope (deferred)

- Animation timeline (movement over time). Deferred to v1.1 or v1.2 based on creator demand signal.
- Multi-sport support. Soccer-only at launch. Field templates and palette items are sport-specific by design; abstracting prematurely will slow v1 without paying off until we have a non-soccer pilot.
- Real-time collaborative editing. Single-author for v1. Sharing is read-only links.
- AI-assisted drill suggestions. Deferred until we have enough drill data to make suggestions non-embarrassing.
- Video-to-diagram extraction. Out of scope at any horizon currently planned.

Requirements

Functional

- **F1.** A coach can create a new drill in under 90 seconds from "new drill" tap to first saved version with at least 4 players and 1 arrow placed.
- **F2.** Field templates: full pitch, attacking half, defending third, midfield third. One-tap switch between them without losing placed objects (objects clamp to the new bounds).
- **F3.** Player tokens render with optional jersey number, color (light/dark/GK), and an optional label. Default is unlabeled — most drills don't need it.
- **F4.** Arrow types: solid (run), dashed (pass), curly (dribble). Arrowhead always indicates direction. No other line styles in v1.

- **F5.** Save creates a versioned snapshot. Editing creates a new version; old versions are recoverable for 30 days. (Rationale: coaches iterate on the same drill across a season; we don't want to lose what worked.)
- **F6.** A drill can be assigned to a player's IDP. Assignment renders the drill image inline in the player app, with the coach's notes as the description.
- **F7.** Tags: free-text, autocompleted from the team's existing tag set. No fixed taxonomy in v1 — let coaches' language emerge before we constrain it.
- **F8.** Export to PNG (web + app) and PDF (web only). PNG is the format the player app and marketplace render against.

Non-functional

- **N1.** Canvas usable on iPad (primary authoring device for coaches). Touch targets ≥44pt; pinch-zoom on the field, not the page.
- **N2.** Drill render in the player app must be ≤500ms on a 4G connection from cold open of the IDP screen.
- **N3.** Drill data stored in a structured JSON format (objects, positions, arrows), not as a flattened image. The PNG is a render artifact; the JSON is the source of truth. This is the decision that keeps v1.1 animation cheap.
- **N4.** Schema versioned from day one. v1 schema is "drill_v1". No migrations promised, but the version field exists so v1.1 can branch cleanly.

Success Metrics

v1 is a question, not a launch. The point of shipping it is to learn whether drills become a marketplace primitive. Metrics laddered to that question:

Activation

- % of pilot coaches who create at least 1 drill in the first 14 days. Target: 60%.
- % of created drills that get assigned to at least 1 player. Target: 50%. (A drill that's built and never assigned is a sketch, not a workflow.)

Engagement

- Median drills created per active coach per week. Target: 3 in steady state. Below 1 means the tool isn't part of session prep; above 5 means it is.
- % of player IDP views in which a drill diagram renders. Target: 40%.

Strategic (the real test)

- Sheldon (and any subsequent marketplace creator) chooses to author at least 1 drill pack natively in the tool, rather than uploading PDFs, within 60 days of v1.1 release. Pass/fail. This is the signal that drills are a marketplace primitive worth investing in further.

Risks & Mitigations

Risk	Likelihood / Impact	Mitigation
Coaches use the tool once, never return — diagram-building isn't a real recurring need.	Medium / High	Activation metric tells us inside 30 days. If <30% of coaches return, kill v1.1 and reallocate.
Players ignore the diagrams; the description text was always sufficient.	Medium / Medium	Track IDP-view-with-diagram engagement vs. without. If no lift, the diagram is decoration, not a feature.
Sheldon prefers his existing PDF/video workflow and doesn't author natively.	Medium / High	Talk to Sheldon before v1.1 ships, not after. If native authoring isn't compelling, the marketplace thesis needs to be reworked.
Animation demand emerges fast and v1's static-only feels broken.	Low / Medium	Schema versioning (N4) keeps the upgrade cheap. Triggered by user feedback, not a roadmap date.
Multi-sport request from a non-soccer pilot before we're ready.	Low / Low	Field rendering is template-driven; adding a sport is a content problem, not an architecture one. Defer until concrete demand.

Open Questions

- **Animation as premium tier?** If v1.1 ships animation, is it a free upgrade or a paid feature for creators? Decision deferred until we see whether creators actually want it.
- **Drill remix rights.** If a coach buys a marketplace drill pack, can they edit it for their team? My current default: yes for personal use, no for redistribution. Needs to be settled before marketplace v1.1.
- **Library scope.** v1 scopes the library to the coach's own team. Open: should there be a public library of community-shared drills, separate from the paid marketplace? Postponed until v2.